

Pranay Dogra

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. in Electrical Engineering & Computer Science and Industrial Engineering & Operations Research

EXPERIENCE

Software Engineer

August 2026 - Present

Space Technologies and Rocketry (STAR), UC Berkeley

Berkeley, CA

- Implemented C++ DAQ server support for an environmental sensor board, parsing UDP packets to ingest real-time temperature, humidity, and pressure telemetry.
- Integrated the board into STAR's configuration, data-streaming, and frontend systems, enabling engineers to monitor environmental conditions through the existing DAQ interface.

Founder

November 2024 - August 2026

CueLens | Featured in MidPeninsula Post

- Developed assistive AI that decodes facial emotions for neurodivergent communication, securing partnerships with three autism clinics across Taiwan, India, and the United States.
- Interviewed 50+ clinicians and users across autism clinics; raised \$5,000 for the Soham Learning Foundation.

Software Engineering Intern

Summers 2024 and 2025

Mirro.ai

- Reduced bias in the DeepSruthi emotion-recognition model by creating a balanced dataset of 40,000+ videos and images.
- Integrated DeepSruthi into multiple robotics applications and created demos for investor meetings.

RESEARCH

Undergraduate Researcher

September 2026 - Present

UC Berkeley Haas School of Business, Dana Carney Lab

Berkeley, CA

- Develop Python and PyTorch pipelines to extract facial, vocal, and paralinguistic features from video and audio for research on political ideology, race, and socioeconomic status.

Student Researcher

June 2023 - August 2023

UC Berkeley AI Research

Berkeley, CA

- Built a 1,020-prompt benchmark evaluating agentic LLM decisions across healthcare, criminal justice, and business scenarios for race, gender, age, socioeconomic status, and political bias.
- Collected choices and logits from FLAN, Vicuna, LLaMA, and LLaMA 2; computed Shannon-entropy bias scores showing FLAN as most biased at 0.602 versus 0.961, 0.976, and 0.965.
- Published "[Quantifying Bias in Agentic Large Language Models: A Benchmarking Approach](#)" at IEEE ICTC 2024.

PROJECTS

Babel | LLMs, Finite State Machines, Multi-Agent Systems

[GitHub](#)

- Built an LLM-powered contract analysis system that converts legal agreements into executable finite state machines, enabling deterministic validation, scenario simulation, and automated execution workflows.
- Designed a multi-agent architecture using GBrain shared memory and validation feedback loops, improving structured FSM generation reliability from 36.9% to 95.3%.

SLM Cascade | Ollama, OpenAI, Claude

[GitHub](#)

- Built a small-model-first inbox agent router using local planning, validation, selective email retrieval, and fallback escalation across Ollama, OpenAI, and Claude.
- Designed a benchmark comparing latency and token usage; reduced token cost by 70%+ with 3.43-second median latency.

TECHNICAL SKILLS

Languages: Java, Python, C, C++

Tools: PyTorch, Git, Ollama, Kubernetes

Systems: UDP Networking, Data Acquisition, Computer Vision, LLM Evaluation and Benchmarking